

Feeding the City: How Rural Productivity Shapes Urban Structural Change

Idaliya Grigoryeva

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PhD Candidate in Economics
UC San Diego

Motivation & research question

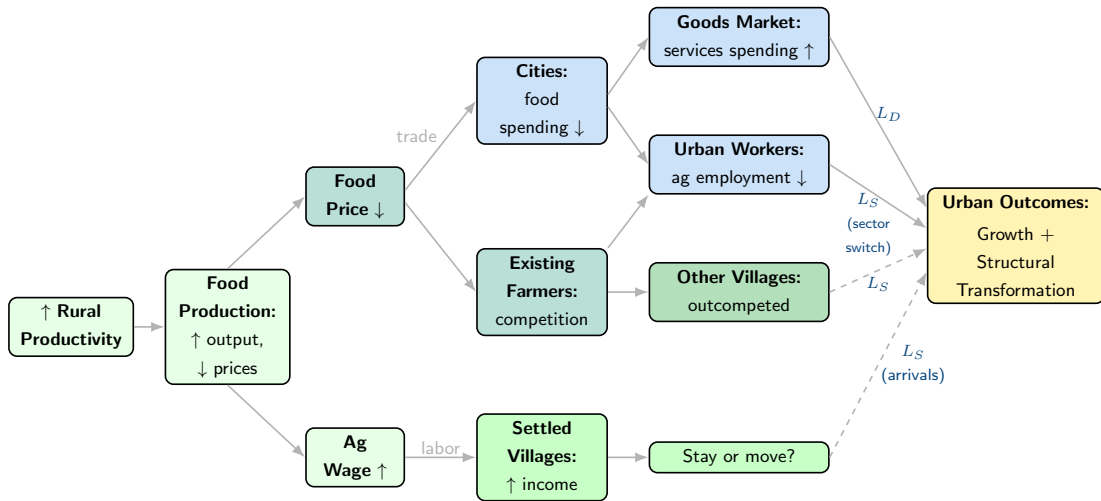
- ▶ Government & development programs often target rural areas while cities offer higher standards of living ($\approx 4\%$ of public spending on agriculture)
- ▶ Classical theories focus on higher ag productivity releasing excess labor to cities, driving industrialization (Harris & Todaro, 1970; Lewis, 1954)
- ▶ Or do productivity gains retain workers in agriculture, lowering incentives to leave? $\Rightarrow$ ambiguous effect (Bustos et al., 2016; Glaeser, 2014; Matsuyama, 1992)
 - ▶ Scarce causal evidence on mechanisms: $\uparrow$ ag productivity $\Rightarrow ? \Rightarrow \uparrow$ urban growth (Bustos et al., 2020; Gollin et al., 2002; Lagakos et al., 2020)
- ▶ **Research Question:**

How does rising agricultural productivity in rural areas affect growth and structural transformation in cities in developing countries?

- ▶ **Setting:** Indonesia's resettlement program as a source of rural productivity variation
- ▶ **Main result:** cities gain manufacturing and services jobs without pop growth
- ▶ **Channels:** rural productivity transforms cities through trade more than migration (data + model)
- ▶ **Contribution:**
 - ▷ rural-to-urban spillovers: village shocks affect city employment ([Bustos et al., 2016](#); [Moretti, 2010](#))
 - ▷ food trade as the channel of a rural shock ([Bryan & Morten, 2019](#); [Tombe & Zhu, 2019](#))

Trade-mig lit

Mechanism map: from rural productivity to city outcomes



The source of the rural productivity shock

↑ Rural
Productivity

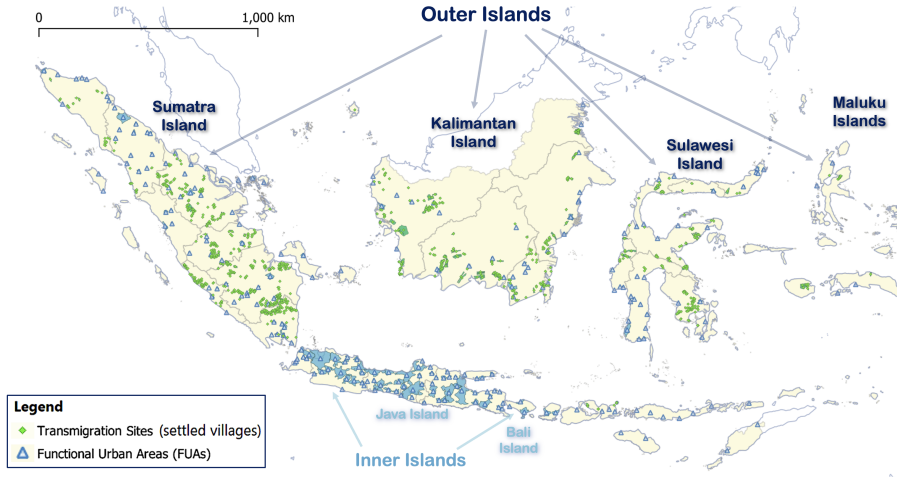
Setting: a rural productivity shock and observed urban outcomes

- ▶ Indonesia's Transmigration (TM) Program resettled 2+ million people in 1979-1988
 - ▶ Free 2-ha land lots with a house + free transportation for families to encourage moving to Outer Islands
 - ▶ Aimed at reducing overpopulation on Java and Bali and improving food security
 - ▶ Migrant destinations were plausibly as good as random
- ▶ Agro-climatic similarity between migrant origins and destinations (soil, climate, geography) is quasi-random & a good proxy for migrant skills and productivity ([Bazzi et al., 2016](#))
- ▶ I construct urban exposure to rural productivity and measure urban outcomes



Sample Destination Village

Setting: transmigration villages scattered across the Outer Islands



[Back](#)

Large variation in agro-climatic conditions in receiving communities

Setting: agro-climatic similarity as a source of rural productivity

Inner Islands

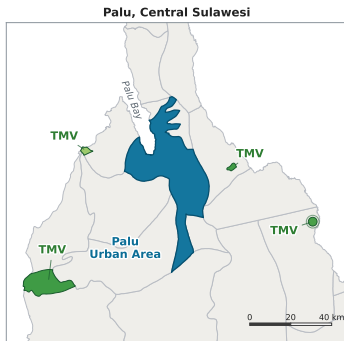


Outer Islands

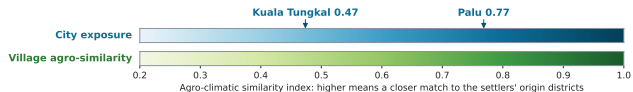
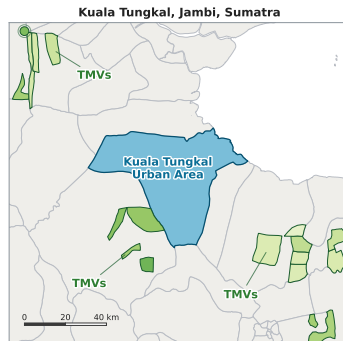


Sample regions: urban exposure to village productivity varies widely

A City with **High** Ag Productivity Shock



A City with **Low** Ag Productivity Shock



Productivity effect *conditional on comparable settler count + agro-climatic village characteristics*

AgSim def

Exposure def

Travel time

Spread

$$UrbanOutcome_c^{2010} = \beta AgSim_c + \gamma Settlers_c + \nu AvgVillageGeo_c + \kappa UrbanBaseline_c^{1980} + \epsilon_c$$

- ▶ β : the causal effect of a productivity shock in the surrounding villages conditional on controls
- ▶ **Identification assumption**: conditional on settler numbers, village agro-climate and the 1980 baselines, match quality is plausibly as good as random
- ▶ Exposure uncorrelated with most 1980 baseline outcomes (unbalanced covar as controls)

AgSim def

Exposure def

Data

Balance

Borusyak-Hull

Result 1: cities change their sectoral employment without pop growth

	Population + Employment			Agric Empl		Modern Empl	
	(1) ln pop	(2) ln emp	(3) unemp rate	(4) ln L agric	(5) agric %	(6) ln L modern	(7) modern %
Exposure to village productivity (1 SD)	0.051 (0.059)	0.024 (0.057)	0.008 (0.007)	-0.166* (0.093)	-0.051*** (0.018)	0.158** (0.062)	0.066*** (0.015)
Transmigrant count (IDW)	✓	✓	✓	✓	✓	✓	✓
Baseline Pop 1980	✓	✓	✓	✓	✓	✓	✓
Urban share 1980	✓	✓	✓	✓	✓	✓	✓
Total Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agri Empl 1980	✓	✓	✓	✓	✓	✓	✓
Agro-Clim. TMV controls	✓	✓	✓	✓	✓	✓	✓
Mean dep var	11.842	10.933	0.122	9.822	0.374	10.232	0.516
Std dev dep var	0.854	0.840	0.040	0.896	0.183	0.975	0.156
Observations	134	134	134	134	134	134	134

⇒ New arrivals cannot account for the shift

How large

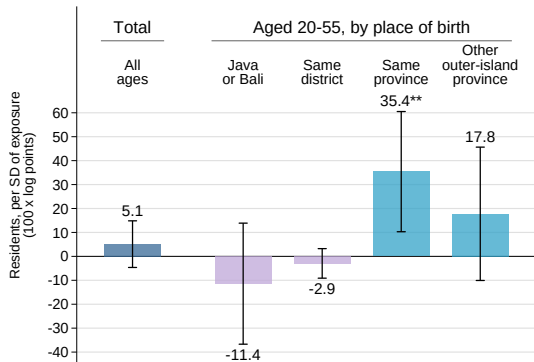
Cohorts

Decomposition

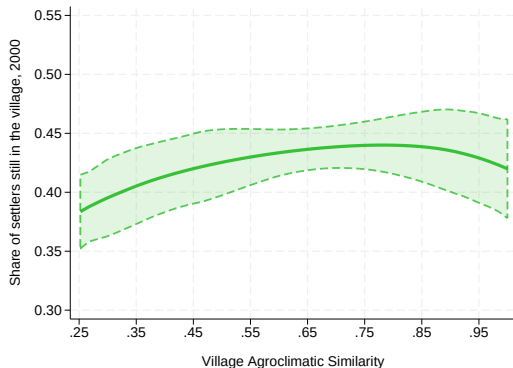
Spec curve

Result 2: transmigrant settlers are not moving differentially

Cities: residents by birthplace (logs)



Villages: share of settlers still there



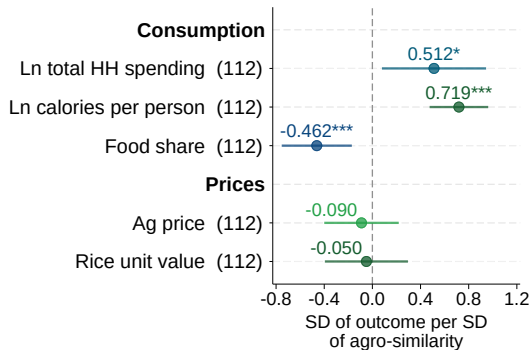
► No rise in city residents born in Java or Bali, the closest proxy for transmigrant settlers

Stay-share

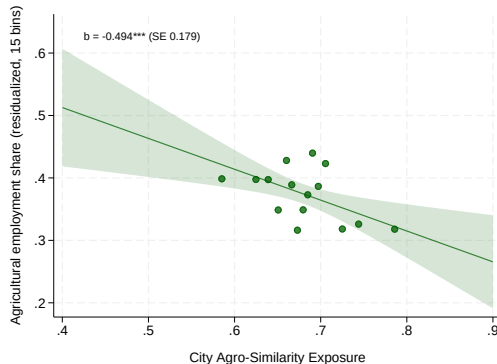
Migration

Result 3: the food expenditure share falls while food consumption does not

Cities: consumption and food prices



Cities: agricultural share *falls*



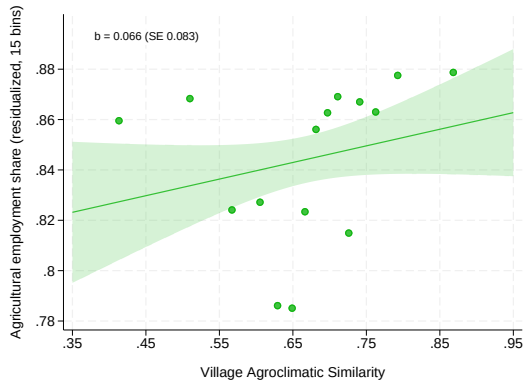
⇒ Spending shifts away from food, freeing up income for other sectors ⇒ $\uparrow L_D$ in non-ag

Budget

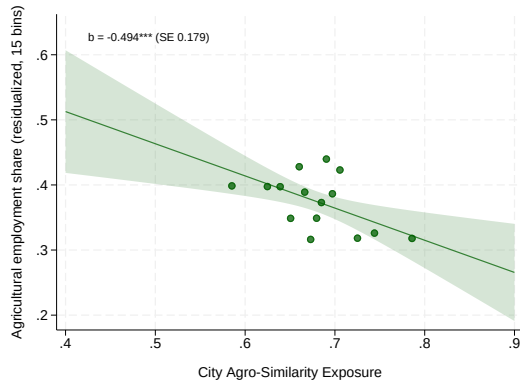
Prices

Result 3: the agricultural share falls in nearby cities and is flat to rising in villages

Villages: agricultural share *flat to rising*



Cities: agricultural share *falls*



The same gain leaves the village agricultural share flat to rising and lowers it in the nearby cities

Semipar view

Result 4: the response is larger in trade-closed cities

A city that can already import food cheaply gains little from a better hinterland ([Glaeser, 2014](#))

	(1) ln pop	(2) ln emp	(3) unemp rate	(4) ln L agric	(5) agric %	(6) ln L modern	(7) modern %
Agro-similarity, more closed cities: β_1	0.118 (0.086)	0.101 (0.082)	0.004 (0.009)	-0.164 (0.132)	-0.065*** (0.024)	0.269*** (0.089)	0.093*** (0.022)
Agro-similarity, more open cities: $\beta_1 + \beta_2$	0.014 (0.059)	-0.018 (0.060)	0.011 (0.007)	-0.170* (0.092)	-0.044** (0.021)	0.099 (0.070)	0.052*** (0.016)
Difference, open – closed	-0.104 (0.075)	-0.119 (0.080)	0.007 (0.009)	-0.007 (0.110)	0.021 (0.025)	-0.169* (0.095)	-0.040* (0.021)
Outcome mean	11.842	10.933	0.122	9.822	0.374	10.232	0.516
Outcome SD	0.854	0.840	0.040	0.896	0.183	0.975	0.156
Observations	134	134	134	134	134	134	134

Note: More open cities = below-median route distance to the Thai and Vietnamese rice hubs.

⇒ **Hinterland productivity matters more for cities with less access to global markets**

(as in the closed economy of [Matsuyama \(1992\)](#))

A two-sector spatial equilibrium of cities and their villages

- **Locations:** Multiple urban and rural regions, each with land and an amenity Model Goals
- **Two sectors:** **food**, traded at a cost rising in travel time, and a **modern sector** (manufacturing + services)
- Workers choose a **location** (elasticity θ_L) first, then a **sector** (elasticity θ_S) Equilibrium
- The model fits the data well Model fit

Stone-Geary Utility

$$U = (c_A - \bar{a})^{\psi_A} c_M^{1-\psi_A}$$

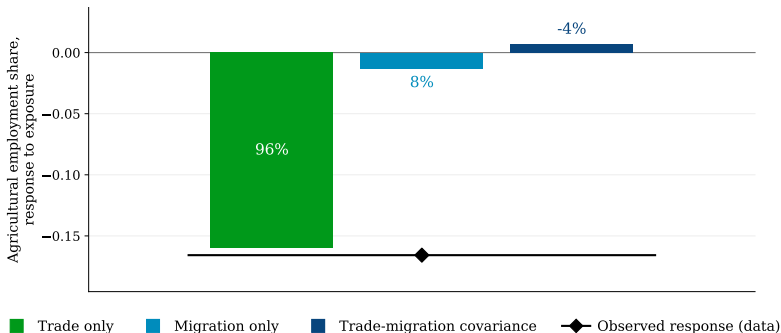
- $\bar{a}$ is a subsistence food floor (enables ag contraction in cities)
- Cheaper food, smaller food share

Stone-Geary

Counterfactuals: the trade-only economy reproduces most of the contraction

- **Goal:** how much of the shift out of agriculture comes from trade, and how much from migration
- **Counterfactuals:** turn off migration/trade $\Rightarrow$ re-solve the model to get new eq'm

How



$\Rightarrow$ **Trade** suffices to explain agricultural contraction

All panels

Rural productivity transforms cities through trade, not migration

- ▶ **Rural-to-urban spillovers exist:** village productivity shifted city employment into manufacturing and services
- ▶ **Modern sector expands,** without changes in population growth or unemployment
- ▶ **Food trade** enables structural change as the food budget share falls, freeing urban workers & spending, especially in cities less connected to global trade
- ▶ $\implies$ Ag productivity gains can translate into expanding services and manufacturing in cities without substantial migration

Thank you!



idagri@ucsd.edu

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